

G Indrasena Yadav

+91 6301440294 | gurram_b230958bt@nitc.ac.in | [GitHub](#) | [LinkedIn](#)

EDUCATION

National Institute of Technology, Calicut

- Bachelor of Technology in Bio Science and Engineering (2023-27) | **CGPA: 7.30** - Kerala
- Narayan Jr College, Nellore | **Percentage: 94.6%** - Andhra Pradesh
- Champion High School, Sullurpeta | **Percentage: 100.0%** - Andhra Pradesh

TECHNICAL SKILLS & TOOLS

Programming with Java, Python

Data Structures and Algorithm's

Web Development:

- HTML, CSS, Java Script
- React, Node.js, Next.js
- SQL, MONGO DB

ML for Data Science & Analytics

- Supervised & Unsupervised Learning Models
- Numpy, Pandas, Matplotlib
- Scikit-Learn
- Tensorflow
- ANN

AWS Cloud Services

- EC2
- Lambda
- DynamoDB

Version Control

- Git
- GitHub

PROJECTS

Study Focus Chrome Extension with Cloud Sync

April-2026

Developed a Chrome Extension with website blocker, Pomodoro focus timer, and study session tracker featuring a personalized user login dashboard.

Implemented cloud sync using REST API and MongoDB to store and sync user preferences, session history, and settings across devices.

Tech Stack:

- **Frontend:** HTML, CSS, JavaScript | **Data Format:** JSON
- **Backend:** REST API (Cloud Sync) | **Database:** MongoDB
- **Platform:** Chrome Extensions API | **Auth:** User Login & Personal Dashboard

Multi-Phase Deep Learning Framework for AMP Classification and Toxicity Prediction

Jan-2026

Developed a multi-phase ML framework for AMP classification, activity prediction, and toxicity analysis using 65K+ peptide sequences, achieving 93%+ accuracy and 0.99 AUC with ensemble learning models.

Built and deployed an end-to-end Streamlit web application integrating ProtT5 embeddings, feature engineering, and CNN-BiLSTM based architectures for real-time peptide prediction and analysis.

Integrated With:

- **ML Models:** SVM, Random Forest, XGBoost, LightGBM, Voting Ensemble
- **DL Models:** CNN, BiLSTM + Attention, Transformer-based Architectures
- **Feature Engineering:** AAC, GAAC, ASDC, Physicochemical Features, ProtT5 Embeddings
- **Data Processing:** CD-HIT Clustering, 5-Fold Cross Validation, External Blind Testing

SIGMA GPT – AI Chatbot Platform (ChatGPT Clone)

May-2026

Developed a full-stack AI chatbot platform with secure user authentication, chat history management, and real-time AI-powered conversations.

Integrated Groq AI API and MongoDB Atlas to provide persistent chat storage, user management, and intelligent responses through a modern web interface

Tech Stack:

- **Frontend:** React.js , vite
- **Backend:** Node.js, Express.js
- **API :-** Groq API key
- **Model :-** Llama 3.3 70B Versatile
- **Database :-** MongoDB
- **Deployment :-** Vercel

SOFT SKILLS & CAMPUS INVOLVEMENT

Soft Skills: Team Collaboration, Innovative, Leadership, Photoshop Design

AI Club: Active member of exploring AI/ML applications

